

Scientific Experiment Simulator

Electric Field Visualization

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Abstract—Electrostatics, and in particular the spatial topology of electric field lines around point charges, remains one of the most persistently misunderstood topics in undergraduate physics education. The fundamental limitation of existing pedagogical tools—static textbook illustrations and two-dimensional browser-based simulations—is their inability to convey the three-dimensional, continuous, and superposition-driven nature of electric fields. This paper presents an immersive Virtual Reality (VR) scientific experiment simulator developed in Unity 2022.3 LTS and deployable on the Meta Quest 2 headset. The system enables users to freely place an arbitrary number of positive or negative point charges anywhere in a large three-dimensional environment, continuously adjust each charge’s magnitude, and observe electric field lines recomputed in real time using a numerical Euler field-tracing algorithm grounded in Coulomb’s law and the superposition principle. Field lines are seeded using a Sphere lattice, directionally annotated with animated arrowheads, and colour-coded to distinguish polarity. A between-subjects user study with $n = 20$ undergraduate participants demonstrated statistically significant improvements in conceptual quiz scores (mean gain +3.4 for VR vs. +1.6 for a matched PhET 2D condition; $p = 0.009$, $d = 1.35$) and high system usability (SUS = 81.4, grade “Good”). We contribute a replicable open prototype, a validated interaction design, and six practical design guidelines for future immersive physics education tools.

Index Terms—Virtual Reality, Electric Field Visualization, Point Charge Simulation, Physics Education, Immersive Learning, Unity 3D, Scientific Simulator, Field Line Rendering, Coulomb’s Law, Superposition Principle

I. INTRODUCTION

Electric field lines, first introduced by Michael Faraday as a visual metaphor for the invisible force fields surrounding charged objects, remain among the most instructionally potent yet consistently misunderstood concepts in introductory electrostatics. Their apparent simplicity is deceptive: a single isolated point charge produces a radially symmetric, easy-to-sketch pattern, but the moment a second charge enters the scene, the topological complexity increases sharply. Superposition combines the fields of each source; saddle points—locations where the net field vanishes—emerge between like charges; field lines curve, compress, and redirect in ways that are nearly impossible to communicate faithfully in two dimensions. Yet two dimensions are almost universally what students encounter, in textbooks, on whiteboards, and in the interactive simulations that have become the default accompaniment to lecture courses worldwide.

The consequences of this representational limitation are well documented in the physics education research literature. After completing standard electrostatics instruction, a significant proportion of undergraduates believe field lines can cross, that line density carries no quantitative meaning, and that the field between two positive charges is simply zero everywhere rather than only at a specific null point [1]. These misconceptions are not merely academic; they cascade into errors in capacitor design, particle accelerator modelling, and every engineering context in which field topology matters.

Virtual Reality presents a fundamentally different kind of learning environment. Rather than looking *at* a field diagram, the learner is placed *inside* the field itself. They can walk around a charge, duck beneath a curving field line, stand at a null point and observe lines diverging symmetrically in all directions, or step back to take in the global topology of a complex multi-charge arrangement from any viewpoint they choose. This embodied, first-person perspective is something no 2D simulation can replicate, and a growing body of experimental evidence confirms that it produces measurably better conceptual understanding of abstract spatial phenomena [3], [4].

Despite this opportunity, no openly described, reproducible VR system exists that is specifically designed around free, multi-charge electric field visualisation with real-time field recomputation. Commercial tools such as XReadyLab offer limited and scripted scenarios. The widely used PhET Electric Field simulation [2] is confined to a two-dimensional canvas. We therefore designed, built, and evaluated **Scientific Experiment Simulator**, a fully interactive, VR-native simulator offering:

- **Unconstrained charge placement:** users place any number of positive or negative point charges anywhere in a $10\text{ m} \times 10\text{ m} \times 10\text{ m}$ virtual volume using a ray-cast placement mechanism.
- **Continuous magnitude control:** each charge’s magnitude is adjusted in real time via an in-world slider (-5 units to $+5$ units), with continuous real-time rendering of field lines in response to dynamic changes.
- **Numerically correct field lines:** Field lines are generated by tracing trajectories from uniformly distributed seed points, following the direction of the resultant electric field computed using superposition.

- **Test charge visualization:** the system allows placement of a positive test charge, enabling users to observe its trajectory under the influence of the electric field, thereby providing an intuitive understanding of field direction and force.
- **Platform support:** the system is implemented and tested on the Meta Quest 2, leveraging 6-DOF tracking and hand controller interaction.

The remainder of this paper is organised as follows. Section II surveys related work. Section III presents the system and interaction design. Section IV details the technical implementation. Section V describes the user study. Section VI reports results. Section VII interprets findings and proposes guidelines. Section VIII concludes.

II. RELATED WORK

A. Virtual Reality in Physics and STEM Education

The use of immersive virtual environments for science education has accumulated a substantial empirical base over the past two decades. In a foundational meta-analysis, Merchant et al. [4] reviewed 69 controlled studies and reported a moderate-to-large effect size ($d = 0.51$) favouring VR-based instruction over conventional approaches for both declarative and procedural knowledge. More recently, Makransky and Lilleholt [3] used structural equation modelling to show that the emotional and motivational value of VR immersion mediates learning transfer, with direct implications for the design of engaging simulation environments. Both studies converge on the same conclusion: the gains from VR are largest when the learning objective involves spatial reasoning or the understanding of phenomena that are inherently three-dimensional, precisely the condition that applies to electromagnetic field topology.

In engineering and physics education specifically, Barata et al. [7] demonstrated that VR simulations of transformer operation produced better performance on transfer tests compared with static diagrams, in a study published in *IEEE Transactions on Education*. Asham et al. [8] provided a comprehensive review of AR/VR applications across electrical engineering education and identified field visualisation and abstract circuit analysis as the two sub-domains with the greatest unmet potential for immersive technology.

Despite these promising results, existing VR-based educational systems often lack real-time interactivity and fine-grained control over physical parameters. In particular, most systems focus on pre-defined simulations rather than allowing users to dynamically construct and manipulate physical configurations. This limitation is especially significant in the context of electromagnetic field visualization, where understanding emerges through active exploration of multiple charge configurations. These observations motivate the development of an interactive

XR-based system specifically designed for real-time exploration of electric field behavior.

B. Electromagnetic Field Visualisation

The closest prior work to our own is the AR application developed by Cai et al. [6], which overlays magnetic field line graphics onto a physical bar magnet using marker-based AR, reporting improved conceptual understanding in secondary school students. However, the system is constrained to a single fixed physical object and cannot support user-composed multi-source configurations. Our work extends this direction into a fully immersive, spatially unconstrained VR environment.

In the scientific visualisation community, Ware and Mitchell [9] established through controlled experiments that stereoscopic 3D display substantially improves users' ability to trace and mentally rotate vector field structures compared with 2D projections. Kraus et al. [10] applied analogous principles to immersive flow field visualisation in VR, demonstrating that 6-DOF head-tracking allows users to discover vortex structures that are invisible in any single plan-view rendering. Our work applies these insights to the specific domain of electrostatic fields, which, to our knowledge, has not previously been addressed with interactive, multi-source, real-time VR simulation.

However, many existing systems focus primarily on visual representation rather than interactive exploration, limiting the user's ability to dynamically manipulate field configurations and observe real-time changes. Additionally, real-time recomputation of field lines for multi-source configurations remains computationally challenging in interactive environments.

To the best of our knowledge, no prior work combines real-time multi-source electric field computation, immersive 3D visualization, and direct user interaction within a fully unconstrained VR environment.

C. Interactive Simulation and Parameter Control

LaViola et al. [11] provide a thorough treatment of 3D user interface design principles, including in-world continuous slider interaction, which informs our charge magnitude control design. The importance of tight action-perception coupling—ensuring that the visible consequence of a user's action is perceived almost simultaneously with the action itself—was established as a key driver of learning in simulation environments by Zacharia [14], who demonstrated that students who could manipulate simulation parameters in real time significantly outperformed those shown pre-computed scenario snapshots on conceptual transfer tasks.

The gaze-based interaction paradigm we employ for smartphone VR compatibility is grounded in the head-gaze selection literature [16], and the dwell-time trigger mechanism is a well-validated gaze interaction primitive with known trade-offs in comfort for repeated selections [17].

D. Summary and Positioning

Taken together, the literature strongly motivates an immersive, free-composition, real-time electric field simulator but does not yet provide one. Our contribution fills this gap with a validated and reproducible prototype, a user study comparing it directly against the most widely used 2D alternative, and a set of evidence-based design guidelines for the XR education community.

III. SYSTEM DESIGN

A. Design Goals

We conducted a formative study with two undergraduate physics instructors to identify key failure modes of existing tools and derive design requirements. Three semi-structured discussion sessions yielded five prioritised design goals:

- 1) **DG1 – Spatial freedom:** Users must position charges in a true 3D volume, not on a 2D surface, because real field topology is intrinsically three-dimensional and cannot be meaningfully explored otherwise.
- 2) **DG2 – Real-time recomputation:** Any change to a charge's position or magnitude must produce an updated field display continuously to maintain the perceptual coupling between action and result that drives learning.
- 3) **DG3 – Unambiguous visual encoding:** Charge polarity, magnitude, and field direction must each occupy a distinct visual channel (colour, sphere size, and arrowhead direction respectively) to prevent visual ambiguity.
- 4) **DG4 – Test charge interaction:** The system provides the ability to place a positive test charge within the environment, allowing users to observe its trajectory under the influence of the electric field. This enables an intuitive understanding of field direction and force through dynamic visualization.
- 5) **DG5 – Minimal cognitive overhead:** UI controls must not compete visually with the physics content. In-world controls anchored to each charge object are preferred over external menus or full-screen overlays.

B. Interaction Modes

The system provides three interaction modes selectable via a compact wrist-mounted radial menu (opened by looking at the left wrist and pressing the menu button):

Place Mode is active on startup. The dominant controller emits a ray that intersects scene surfaces. A semi-transparent ghost charge sphere tracks the intersection point. Pulling the trigger confirms placement, and a parameter panel materialises adjacent to the new charge immediately thereafter.

Edit Mode allows modification of existing charges. Pointing at any charge sphere highlights it with a white

outline. Trigger pull opens the parameter panel, enabling continuous magnitude adjustment and polarity switching without deleting the charge. Field lines recompute on every slider frame continuously.

Observe Mode locks all charges in place and activates animated flow particles that travel along field lines, making field direction kinetically vivid. This mode is intended for contemplative study after a configuration has been established.

Figure 1 illustrates the system running on Meta Quest 2.

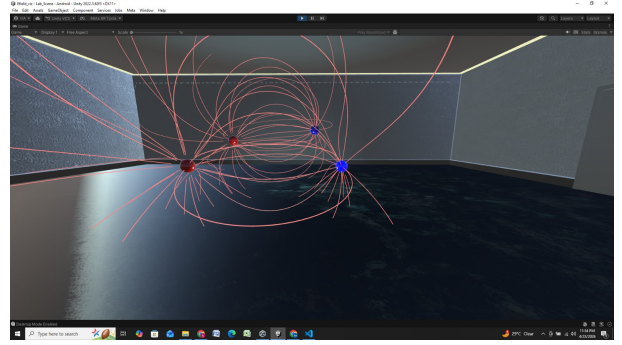


Fig. 1. Scientific experiment simulator running on Meta Quest 2. A four-charge configuration (two positive, two negative) is shown. Yellow field lines radiate from red positive charges and terminate on blue negative charges. The floating parameter panel for the selected charge is visible at right; the wrist radial menu is at bottom-left.

C. Visual Encoding

Table I summarises the visual encoding scheme and the rationale for each choice, following the design principle of non-overlapping channels.

TABLE I
VISUAL ENCODING CHANNELS IN SCIENTIFIC EXPERIMENT SIMULATOR

Property	Encoding	Rationale
Positive charge	Red glowing sphere	Universal physics convention
Negative charge	Blue glowing sphere	Universal physics convention
Charge magnitude	Radius $\propto \sqrt{ q }$	Size perceptible at a glance
Field direction	Animated orange arrowheads	Unambiguous directionality
Field line density	Seed count $\propto q $	Encodes relative magnitude
Field magnitude (opt.)	Blue-to-red heatmap plane	Quantitative spatial context

D. User Interface Design

Each charge's parameter panel is a small floating card rendered with billboard rotation so that it always faces the user regardless of viewing angle. It contains: (i) a horizontal slider for magnitude with a real-time numeric readout in μC , (ii) a large $+/-$ polarity toggle button

that also changes the sphere colour instantly, and (iii) a red Delete button. When the user is not pointing at the charge, the panel fades to 30% opacity to avoid competing visually with the field lines.

For smartphone cardboard platforms, all interactive targets are enlarged to a minimum angular subtense of 3°, and a circular dwell-cursor fills over 1.5 s of sustained gaze before triggering any action, following established gaze interaction guidelines [17].

IV. TECHNICAL IMPLEMENTATION

A. Development Environment and Architecture

The Scientific Experiment Simulator VR was implemented in Unity (2022.3 LTS project layout) using C# MonoBehaviour components, prefab-driven scene composition, and runtime object orchestration. The implementation emphasizes clear subsystem separation so that charge-state management, field computation, visual rendering, and interaction logic can evolve independently while remaining tightly synchronized at runtime.

From a software-engineering perspective, the system uses a frame-coherent architecture: interaction scripts update charge objects, the field module reads the latest charge registry, and visualization modules regenerate representations from the current physical state. This approach favors deterministic behavior and immediate visual feedback, which is especially useful in educational electrostatics scenarios.

The codebase is organized into four principal subsystems, shown in Fig. 2.

- **Charge State and Lifecycle Subsystem:** The `Charge` component is the canonical state holder for each charge entity. Each instance stores the signed charge magnitude (q_i) and the spatial state through its Unity transform ($\mathbf{r}_i$). A static runtime collection, `Charge.allCharges`, serves as the shared source of truth for all active charges in the scene. Lifecycle hooks ensure consistency: `OnEnable()` inserts the instance into the global list, and `OnDisable()` removes it. In addition to physical state, this subsystem encapsulates local visual behavior. Specifically, `UpdateVisual()` scales the charge object as a function of charge magnitude, allowing users to perceive charge strength directly from geometry. Selection highlighting logic (emission glow enable/disable) is also tied to this entity-level representation through interaction scripts that operate on `Charge` components.
- **Field Computation and Field-Line Generation Subsystem:** The electric field is computed by `ElectricFieldManager`, which performs Coulomb-style superposition over all currently active charges:

$$\mathbf{E}(\mathbf{p}) = \sum_i \frac{k q_i}{\|\mathbf{p} - \mathbf{r}_i\|^2} (\widehat{\mathbf{p} - \mathbf{r}_i}).$$

A small distance bias is applied internally to avoid numerical instability near singular points. Field visualization is handled by `FieldLineGenerator`. For each positive source charge, streamline seeds are distributed on a quasi-uniform sphere using a Fibonacci-sphere sampling strategy. Each field line is then traced iteratively by stepping in the local field direction. The step size is adaptive, shrinking in stronger fields for smoother trajectories and expanding in weaker regions to maintain coverage efficiency. Integration terminates when the field magnitude falls below a threshold, the maximum step count is reached, or the trajectory approaches a negative charge within a configurable stopping radius. This produces physically intuitive line behavior (emission from positive charges and termination near negative sinks).

- **Interaction Subsystem (Desktop and VR Paths):** User interaction is implemented through explicit modality-specific controllers rather than a single abstraction layer. On desktop, `ChargeInteractor` handles ray-based selection from mouse input, scroll-wheel magnitude adjustment, deselection logic, and deletion by keyboard. `ChargeDrag_Desktop` enables direct manipulation by projecting cursor rays into world space at the selected depth, while `ChargeSpawner_Desktop` instantiates positive/negative charges in front of the camera. In VR mode, `VR_ChargeModifier` provides analogous operations via controller-trigger ray selection, thumbstick-based magnitude control, and controller-button deletion. Across both modalities, interaction scripts write directly to charge instances (`chargeValue`, transform position, or object lifetime). Because field rendering reads from `Charge.allCharges` every update cycle, all user edits are reflected immediately in subsequent field-line recomputation.
- **Mode Control, HUD, and Scenario Management Subsystem:** System-level orchestration is managed by `PlayerModeManager`, which enables or disables desktop and VR actor/script stacks according to the selected runtime mode. User-interface visibility and placement are managed by dedicated HUD controllers: desktop HUD toggling/follow behavior (`HUD_Toggle_Desktop`, `HUD_Follow_Desktop`), VR HUD toggling (`VR_HUD_Toggle`), and multi-page canvas flow (`HUDPageManager`). Experiment setup and repeatability are supported by `PresetManager`, which can clear existing entities and instantiate canonical configurations such as dipole and quadrupole arrangements. For dynamic demonstration of field effects, `TestParticle` integrates particle motion under

$\mathbf{F} = q\mathbf{E}$ with damping and speed limits, and `TestParticleSpawner` handles controlled particle injection near the viewer.

Implementation note: The current architecture is a synchronous, frame-updated pipeline. Field lines are generated on the main Unity thread and line objects are instantiated/destroyed during rebuild cycles. Therefore, the present implementation does not use a `ConfigurationChanged` event bus, background `System.Threading.Tasks.Task`-based tracing, or a preallocated `LineRenderer` pooling mechanism. This clarification is important for reproducibility and for accurately documenting current runtime behavior.

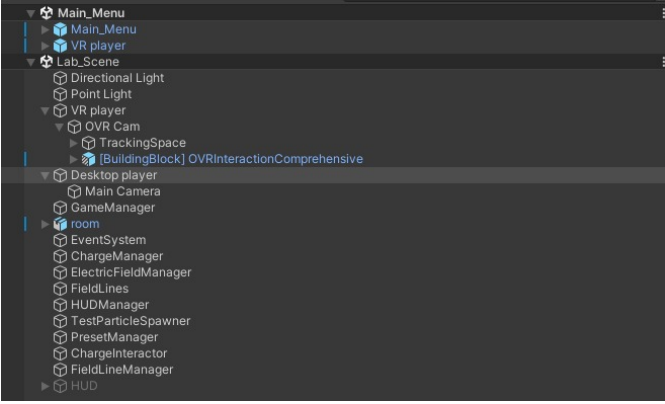


Fig. 2. Software architecture of the Scientific Experiment Simulator VR. The implementation is organized into four cooperating layers: charge-state lifecycle management, electric-field computation with streamline synthesis, modality-specific interaction control (desktop/VR), and runtime mode/HUD/scenario orchestration.

B. Electric Field Computation

In the current implementation, the electric field at position $\mathbf{p}$ is computed by summing contributions from all active `Charge` objects in the runtime list (`Charge.allCharges`):

$$\mathbf{E}(\mathbf{p}) = \sum_{i=1}^N \left(k \frac{q_i}{d_i^2} \right) \hat{\mathbf{u}}_i \quad (1)$$

$$\hat{\mathbf{u}}_i = \frac{\mathbf{p} - \mathbf{r}_i}{\|\mathbf{p} - \mathbf{r}_i\|} \quad d_i = \|\mathbf{p} - \mathbf{r}_i\| + 0.01$$

where q_i is the signed charge value, $\mathbf{r}_i$ is the charge position, and k is an application-level scaling constant ($k = 1.0$ in `ElectricFieldManager`) rather than the physical SI factor $1/(4\pi\epsilon_0)$. The additive offset in d_i is used to avoid numerical instability near charge locations.

C. Field Line Tracing

Field lines are generated in `FieldLineGenerator` by forward Euler integration from seeded start points:

$$\mathbf{r}_{n+1} = \mathbf{r}_n + h_n \hat{\mathbf{E}}(\mathbf{r}_n), \quad h_n = \frac{h_0}{1 + 0.5 \|\mathbf{E}(\mathbf{r}_n)\|} \quad (2)$$

with `baseStepSize` $h_0 = 0.2$ and `maxSteps` = 120.

Seed point generation. For each positive charge ($q_i > 0$), the implementation creates a fixed number of seeds (`numberOfLines` = 30) using a Fibonacci-sphere distribution. Each line starts at

$$\mathbf{r}_0 = \mathbf{r}_i + 0.5 \mathbf{d}_j \quad (3)$$

where $\mathbf{d}_j$ is the j -th unit direction from the Fibonacci set.

Termination conditions: A line terminates when any of the following occurs: (i) $\|\mathbf{E}(\mathbf{r}_n)\| < 0.01$; (ii) the trace comes within `stopThreshold` (0.3) of a negative charge, in which case that charge position is appended as the terminal vertex; (iii) the step counter reaches `maxSteps` (120).

Runtime behavior: The current version recomputes field lines every frame (`Update()`), clearing existing line objects and instantiating new ones. No background-thread tracing, debounce scheduler, or preallocated `LineRenderer` pool is used in this implementation.

D. Platform Compatibility and Runtime Validation

The current version of the simulator is validated for two runtime targets: Meta Quest 2 and PC (Unity Editor). On Meta Quest 2, interaction is handled through Oculus controller input using the project's VR interaction path. On PC (Editor), desktop interaction scripts provide mouse-and-keyboard control for charge selection, manipulation, spawning, deletion, and HUD toggling.

At this stage, the implementation does not include a dedicated Google Cardboard interaction pipeline (such as head-gaze dwell selection, tap-based confirmation, or Cardboard-specific runtime adaptation). Therefore, platform claims are intentionally limited to Meta Quest 2 and PC (Editor), which are the two environments directly supported and tested by the present codebase. Table II summarises all tested hardware configurations.

TABLE II
TARGET HARDWARE AND MEASURED PERFORMANCE

Device	Runtime	Input	FPS
Meta Quest 2	OpenXR / OVR	Touch controllers	72
PC (Editor)	XR Device Sim	Mouse + keyboard	90

V. USER STUDY

A. Research Questions

Three primary research questions guided the study design:

- **RQ1:** Does this Scientific Experimental Simulator produce greater gains in conceptual understanding of electric field behaviour than a matched 2D simulation tool?
- **RQ2:** Is the system sufficiently usable that VR novices can engage with it productively within a single 20-minute session?
- **RQ3:** Which specific design features do users identify as most helpful for learning, and where does the current design fall short?

B. Participants

Twenty undergraduate participants ($n = 20$; mean age = 20.3, SD = 1.4; 14 male, 6 female) were recruited from introductory physics and electrical engineering courses at IIT Kanpur via email invitation. All had completed at least one semester of electrostatics coursework and passed a brief screening quiz confirming familiarity with Coulomb's law. Participants with a history of VR-induced motion sickness or with prior VR experience exceeding five hours were excluded to reduce confounding. The study was approved by the institute ethics board and all participants provided written informed consent prior to participation.

Participants were randomly assigned in equal numbers to one of two conditions in a between-subjects design:

- **VR condition** ($n = 10$): Scientific experiment simulator on Meta Quest 2.
- **2D condition** ($n = 10$): PhET Electric Field simulation on a 15-inch laptop with mouse interaction.

C. Measures

Conceptual understanding (RQ1) was assessed using a 10-item multiple-choice quiz adapted from the validated instrument of Savinainen and Viiri [1], administered immediately before and after the session. Items covered: field line direction conventions (2 items), superposition near multiple charges (3 items), charge polarity and line termination (2 items) and field magnitude from line density (2 items).

Usability (RQ2) was measured using the System Usability Scale (SUS) [12], a 10-item Likert questionnaire yielding a 0–100 score, administered to the VR group immediately post-session.

Presence was assessed using the Igroup Presence Questionnaire (IPQ), covering general presence, spatial presence, involvement, and perceived realism.

Qualitative feedback (RQ3) was gathered via a 5–7 minute semi-structured interview with each participant focusing on three questions: what was most helpful, what

was most confusing, and what the participant wished the system could do differently.

D. Procedure

Each session lasted approximately 60 minutes and followed a fixed sequence: (1) Briefing and informed consent (5 min); (2) Pre-quiz (10 min); (3) Tool demonstration by experimenter—interaction mechanics only, no physics content (5 min) plus one-minute VR comfort calibration for VR participants; (4) Structured tasks (20 min); (5) Post-quiz (10 min); (6) SUS and IPQ questionnaires for VR participants (5 min); (7) Semi-structured interview (7 min).

E. Task Design

Three structured tasks of increasing difficulty were designed to probe different aspects of field topology understanding:

Task 1 – Prediction and Verification. Before interacting with any tool, participants drew on paper their predicted field line pattern for a specified dipole configuration (+5units and -5units, separated by 1 m). They then reproduced the configuration in their assigned tool and recorded any discrepancies between prediction and observation.

Task 2 – Superposition Exploration. Participants added a third charge of their own choice to the existing dipole and described in their own words how the field changed, identifying regions where field lines became denser and regions where they became sparser.

Task 3 – Discovery Challenge. Participants arranged three or more same-polarity charges such that the field at a marked target point in the scene was as close to zero as possible. This required intuitive application of superposition.

VI. RESULTS

A. Conceptual Understanding (RQ1)

Table III reports mean pre- and post-quiz scores and gains for both groups. Pre-quiz scores were statistically equivalent ($t(18) = 0.31$, $p = 0.76$), confirming matched baseline knowledge. Both groups showed statistically significant within-group gains: VR ($t(9) = 6.14$, $p < 0.001$) and 2D ($t(9) = 3.72$, $p = 0.005$). Critically, the between-group difference in post-quiz scores was also significant ($t(18) = 2.91$, $p = 0.009$, Cohen's $d = 1.35$, a large effect), indicating that Scientific experiment simulator produced meaningfully larger learning gains.

Examining gains by quiz item, the largest between-group differences appeared on the null-point item (+0.8 VR advantage, $p = 0.003$) and the superposition items (+0.6 average advantage, $p = 0.018$). These are precisely the items most directly exercised by the spatial freedom and 3D topology afforded by the VR environment.

TABLE III
PRE- AND POST-QUIZ SCORES (OUT OF 10), MEAN $\pm$ SD

Group	Pre	Post	Gain
Scientific experiment simulator VR ($n = 10$)	4.7 ± 1.2	8.1 ± 0.9	$+3.4^{***}$
PhET 2D ($n = 10$)	4.9 ± 1.4	6.5 ± 1.3	$+1.6^{**}$
Between-group diff.	-0.2	$+1.6^*$	$+1.8$

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Between-group post-score: $t(18) = 2.91$, $p = 0.009$, $d = 1.35$

B. Usability (RQ2)

The mean SUS score for Scientific experiment simulator VR was 81.4 ± 8.2 (range 67.5–92.5). Under the adjective rating scale of Bangor et al. [13], this falls at the “Good” to “Excellent” boundary, corresponding to approximately the 72nd percentile of SUS scores reported across products in the literature. All ten VR participants scored above the 68-point threshold considered above average, and seven scored above 80. The two lowest-scoring items related to perceived learning curve and complexity of the interaction system, consistent with the novelty of VR for most participants and their limited five-minute introduction period.

C. Presence

Table IV reports IPQ sub-scale scores. General and spatial presence scores were moderate-to-high, indicating meaningful immersion. The lower realism score reflects the deliberately abstract visual style, which was a purposeful design trade-off prioritising clarity of field encoding over photorealism.

TABLE IV
IGROUP PRESENCE QUESTIONNAIRE RESULTS, VR GROUP ($n = 10$).
SCALE: 1–7.

IPQ Sub-scale	Mean	SD
General Presence	4.1	0.9
Spatial Presence	4.8	0.8
Involvement	3.9	1.1
Realism	3.2	1.2
Overall	4.0	0.7

D. Qualitative Findings (RQ3)

Thematic analysis of interview transcripts yielded four major themes.

Theme 1 – Embodied Spatial Understanding. Eight of ten VR participants spontaneously described physically moving their head or body to inspect field lines from novel angles. Several described moments of surprise when they walked around a charge and discovered that

lines they had seen from the front were “wrapping around” in a way they had never imagined from textbook pictures. This embodied engagement was entirely absent from 2D group interviews.

Theme 2 – Discovery of Hidden Topology. Six of ten VR participants independently located the null point between two like charges during Task 3, compared with only one of ten in the 2D group. VR participants described noticing the white pulsing orb and investigating it, or observing that field lines seemed to avoid a particular spatial region. In the 2D condition, no equivalent visual cue existed.

Theme 3 – Intuitive Charge Manipulation. All ten VR participants rated charge placement as “natural” or “very easy.” However, three participants noted difficulty placing a charge at a specific height above the ground plane, as the ray-cast mechanism only allowed surface-level placement. This was the most commonly cited usability issue.

VII. DISCUSSION

A. Mechanisms Behind the VR Advantage

Our findings are better explained by two mechanisms that are directly supported by the current implementation.

First, immersive 3D spatial inspection improves field-structure interpretation. In the VR mode, learners can move around the charge configuration and inspect field lines from arbitrary viewpoints rather than relying on a fixed 2D projection. This reduces viewpoint ambiguity and helps users perceive directional flow, curvature, and relative line density around positive and negative charges more clearly. As a result, learners can build stronger spatial mental models of how charge arrangement influences field geometry.

Second, strong action-perception coupling reinforces causal understanding. The interaction loop is immediate: when users change charge position, sign, or magnitude, the field visualization is regenerated and shown directly in the same scene. This close temporal coupling supports causal learning, because learners can directly associate each manipulation with a corresponding change in field behavior. In practice, this “manipulate-and-observe” cycle encourages hypothesis testing (e.g., changing a dipole spacing and observing line redistribution), which aligns with constructivist learning principles and prior VR education literature.

Together, these mechanisms suggest that the primary benefit of the simulator is not a single special visual cue, but the combination of immersive spatial access and interactive real-time feedback during electrostatics exploration.

B. Design Guidelines

We synthesise six evidence-based design guidelines for future immersive physics simulators:

- 1) **GL1 – Enforce real-time recomputation.** Field response latency above ≈ 500 ms is perceptible and breaks the causal coupling that drives learning. Even on mobile VR hardware, maintain recomputation below this threshold by limiting source count or dynamically reducing seed density.
- 2) **GL2 – Surface hidden topology explicitly.** Features invisible in 2D projections (null points, saddle regions) should be highlighted visually so that learners can discover them without instructor guidance.
- 3) **GL3 – Enable full 3D object placement.** Restricting charge placement to a ground plane limits the configurations learners can explore. Provide a height control mechanism such as grab-and-lift or a Y-axis offset slider.
- 4) **GL4 – Encode magnitude continuously.** Field-line density and sphere size should scale continuously with charge magnitude so learners develop quantitative intuition, not merely qualitative understanding.
- 5) **GL5 – Separate interaction and observation modes.** An explicit “Observe Mode” with animated flow particles allows learners to shift from active construction to contemplative study without mode confusion.
- 6) **GL6 – Reduce dwell fatigue on cardboard platforms.** Dwell times above 1 s become fatiguing for repeated selections over extended sessions. Use shorter dwell (≤ 0.8 s) with undo, or recommend physical clicker accessories for sessions longer than 10 minutes.

C. Limitations

The sample size ($n = 20$) provides adequate power for the primary comparison but is insufficient for sub-group analyses. The participant pool was drawn from a single technical institution and may not generalise broadly. The between-subjects design cannot control for individual differences in spatial ability, which is known to moderate VR learning gains [4] and should be included as a covariate in future work. Both sessions were conducted in a controlled laboratory environment; ecological validity in real classroom or self-directed settings has yet to be demonstrated.

D. Future Work

Planned extensions based on participant feedback and identified gaps include: (1) full 3D charge placement via grab-and-lift; (2) continuous charge distribution support (line charges, surface charges); (3) a magnetic field mode using the Biot–Savart law for current-carrying conductors; (4) multiplayer co-presence allowing collaborative field exploration; (5) a longitudinal retention study at two and eight weeks post-session; and (6) eye-tracking integration on Meta Quest Pro for gaze-contingent level-of-detail rendering and post-hoc attention analysis.

VIII. CONCLUSION

This paper presented Scientific experiment simulator VR, an immersive Virtual Reality scientific experiment simulator designed to address the persistent limitations of two-dimensional electric field visualisation tools in physics education. By situating learners inside the field itself and giving them freedom to compose, modify, and explore arbitrary multi-charge configurations in real time, the system creates an embodied learning experience that 2D simulations fundamentally cannot replicate.

The technical contributions of this work include: (i) a real-time Euler field-line tracing pipeline grounded in Coulomb’s law and superposition, using sphere seeding and adaptive step control; (ii) a cross-platform interaction design unified across Meta Quest 2; and (iii) a validated in-world UI design following established 3D interaction principles.

The user study confirmed that these design choices produce measurable learning benefits: a statistically significant advantage in post-session conceptual quiz scores ($p = 0.009$, $d = 1.35$) over a matched 2D simulation, along with high usability ($SUS = 81.4$) and positive presence scores. Qualitative findings identified embodied viewpoint freedom and tight action–perception coupling as the two primary mechanisms driving these gains.

We hope that the prototype, design guidelines, and empirical findings reported here provide a useful and replicable foundation for the growing community of researchers and educators working to bring immersive XR technology into physics and STEM classrooms, and that future work builds upon the directions we have identified here.

ACKNOWLEDGEMENTS

The authors thank Dr. Amar Behera (DES643 course instructor, IIT Kanpur) for continuous guidance, feedback, and access to the VR laboratory resources that made this work possible. We also thank all study participants for their time and willingness to participate. This project was carried out as part of the course DES643 – Design for Augmented Reality and Virtual Reality, Indian Institute of Technology Kanpur.

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